Notes

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0.1 Complexes and exact sequences

Definition 0.1. A **chain complex** $(X_n, d_n : X_n \to X_{n-1})_{n\in\mathbb{Z}}$ is a sequence of acts and act morphisms

$$\dots \longrightarrow X_{n+1} \xrightarrow{d_{n+1}} X_n \xrightarrow{d_n} X_{n-1} \longrightarrow \dots$$

such that for all $n \in \mathbb{Z}$ we have $\mathrm{im}(d_{n+1}) \subseteq \ker(d_n)$. Chain complex is **Rees exact at** X_n if and only if $\mathrm{im}(d_{n+1}) = \ker(d_n)$. Chain complex is **Rees exact** if it is Rees exact at X_n for each $n \in \mathbb{Z}$.

Remark. If $(X_n, d_n : X_n \to X_{n-1})_{n\in\mathbb{Z}}$ is a sequence of pointed acts, then it is a chain complex if and only if $\forall n \in \mathbb{Z} : d_n \circ d_{n+1} = 0$.

Proof. First we assume that $d_{n+1} \circ d_n = 0$. Let $(y_1, y_2) \in \mathrm{im}(d_{n+1})$, if $y_1 = y_2$ we are done. Suppose then that there are some $x_1, x_2 \in X_{n+1}$ such that $d_{n+1}(x_1) = y_1$ and $d_{n+1}(x_2) = y_2$, then we have

$$d_n(y_1) = (d_n \circ d_{n+1})(x_1) = 0 = (d_n \circ d_{n+1})(x_2) = d_n(y_2)$$

therefore $(y_1, y_2) \in \ker(d_m)$, hence $\mathrm{im}(d_{n+1}) \subseteq \ker(d_n)$.

Assume now that $\mathrm{im}(d_{n+1}) \subseteq \ker(d_n)$. We know that $0 \in \mathrm{im}(d_{n+1})$, since $d_{n+1}(0) = 0$ therefore for any $x \in X_{n+1}$ by assumption $(0, d_{n+1}(x)) \in \ker(d_n)$ so $d_{n+1}(d_n(x)) = d_n(0) = 0$. $\square$

Definition 0.2. Let $(X_n, d_n)_{n\in\mathbb{Z}}$ and $(Y_n, e_n)_{n\in\mathbb{Z}}$ be two chain complexes. A **chain map** between them is a sequence of act morphisms $(f_n : X_n \to Y_n)_{n\in\mathbb{Z}}$ such that for all $n \in \mathbb{Z}$ the following square commutes

$$\begin{tikzcd}
X_{n+1} \arrow[r, "d_{n+1}"] \arrow[d, "f_{n+1}"'] & X_n \arrow[d, "f_n"] \\
Y_{n+1} \arrow[r, "e_{n+1}"] & Y_n
\end{tikzcd}$$

Remark. We have that $f_n(\mathrm{Im}(d_{n+1})) \subseteq \mathrm{Im}(e_{n+1})$.

Definition 0.3. Let $(X_n, d_n : X_n \to X_{n-1})_{n\in\mathbb{Z}}$ be a chain complex, then we define **the n-th homology act** by

$$h_n(X) = \ker(d_n)/\mathrm{im}(d_{n+1}) = \{[x, y] : (x, y) \in \ker(d_n)\}$$

where by $\ker(d_n)/\mathrm{im}(d_{n+1})$ we mean the Rees factor by a subact.

Let (A, a) and (B, b) be chain complexes and $f : A \to B$ a chain map, then define $h_n(f) : h_n(A) \to h_n(B)$ by $[x, y] \mapsto [f_n(x), f_n(y)]$. It is well-defined, since if $(x, y) \in \ker(a_n)$, then $b_n(f_n(x)) = f_{n-1}(a_n(x)) = f_{n-1}(a_n(y)) = b_n(f_n(y))$. And if we choose $[x_1, y_1] = [x_2, y_2]$, then either $(x_1, y_1) \sim (0, 0) \sim (x_2, y_2)$ or $x_1 = x_2$ and $y_1 = y_2$. If the latter then we are done, otherwise it suffices to show

$$[(x_1, y_1) \sim (0, 0) \iff (x_1, y_1) \in \mathrm{im}(a_{n+1})] \implies (f_n(x_1), f_n(y_1)) \sim (0, 0)$$

If $x_1 = y_1$, then $f_n(x_1) = f_n(y_1)$ which implies $(f(x_1), f(y_1)) \sim (0, 0)$. Otherwise we find $x_2, y_2 \in A_{n+1}$ such that $a_{n+1}(x_2) = x_1$ and $a_{n+1}(y_2) = y_1$, but then $f_n(x_1) = f_n(a_{n+1}(x_2)) = b_{n+1}(f_{n+1}(x_2))$ and $f_n(y_1) = f_n(a_{n+1}(y_2)) = b_{n+1}(f_{n+1}(y_2))$, therefore $(f_n(x_1), f_n(y_1)) \in \mathrm{im}(b_{n+1}) \iff (f_n(x_1), f_n(y_1)) \sim (0, 0)$.

Proposition 0.1. *$h_n : Ch(\mathcal{S} - \boldsymbol{Act}) \to \mathcal{S} - \boldsymbol{Act}$ is a functor.*

Proof.

$$h_n(1_A)([x,y]) = [1_{A_n}(x), 1_{A_n}(y)] = [x,y] \implies h_n(1_A) = 1_{h_n(A)}$$

$$h_n(g \circ f)([x,y]) = [g_n(f_n(x)), g_n(f_n(y))] = h_n(g)([f_n(x), f_n(y)]) = (h_n(g) \circ h_n(f))([x,y])$$

therefore $h_n(g \circ f) = h_n(g) \circ h_n(f)$. □

From now on by acts we will mean pointed acts, objects of the category $\mathcal{S} - \mathbf{Act}_0$.

Definition 0.4. Rees short exact sequence is a sequence of acts

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

exact at A, B and C.

Remark. By definition of exactness it follows that $\ker(f) = \{(0,0)\} \cup \Delta_L$ and $\mathrm{im}(g) = N \times N$, therefore f is injective and g is surjective. Similarly injectivity of f and surjectivity of g imply exactness at A and C respectively.

Example 0.1. Let A and B be acts, then we have the following sequence

$$0 \longrightarrow A \xrightarrow{\iota} A \oplus B \xrightarrow{\lambda} B \longrightarrow 0$$

where $\iota : x \mapsto (x, 0)$ and

$$\lambda(x,y) = \begin{cases} y, & \text{if } x = 0 \\ 0, & \text{otherwise} \end{cases}$$

the exactness at A and B is easily seen. We will show exactness at $A \oplus B$.

Choose $((x_1,y_1),(x_2,y_2)) \in \mathrm{im}(\iota)$, if $x_1 = x_2$ and $y_1 = y_2$ we are done, otherwise we have $y_1 = 0 = y_2$, therefore

$$\lambda(x_1,y_1) = 0 = \lambda(x_2,y_2)$$

which implies $\mathrm{im}(\iota) \subseteq \ker(\lambda)$. Let $((x_1,y_1),(x_2,y_2)) \in \ker(\lambda)$, if $x_1 = x_2$ and $y_1 = y_2$ we are done, therefore assume that $x_1 \neq x_2$ or $y_1 \neq y_2$.

Suppose $x_1 = 0$, then $x_2 \neq 0$ (else $y_1 = y_2$), this implies that $y_2 = 0$ and $y_1 = \lambda(x_1,y_1) = \lambda(x_2,y_2) = 0$ so $(x_1,y_1) = (0,0)$ and $y_2 = 0$, then $\iota(x_2) = (x_2,y_2)$ and $\iota(0) = (x_1,y_1)$.

Suppose now that $x_1 \neq 0$, then $y_1 = 0$ and $\lambda(x_2,y_2) = \lambda(x_1,y_1) = 0$, which means either $(x_2,y_2) = (0,0)$ or that $x_2 \neq 0$, in either case $y_2 = 0$, hence $\iota(x_1) = (x_1,y_1)$ and $\iota(x_2) = (x_2,y_2)$. In conclusion $\ker(\lambda) = \mathrm{im}(\iota)$ as required.

Lemma 0.1. *Consider the following commutative diagram*

$$\begin{tikzcd}
0 \arrow[r] & A \arrow[r, "f"] \arrow[d, "\alpha"] & B \arrow[r, "g"] \arrow[d, "\beta"] & C \arrow[r] \arrow[d, "\gamma"] & 0 \\
0 \arrow[r] & A' \arrow[r, "f'"] & B' \arrow[r, "g'"] & C' \arrow[r] & 0
\end{tikzcd}$$

where α, β, γ are isomorphisms, then the first row is Rees exact iff the second row is.

Proof. Assume that the first row is Rees exact. We will prove that f' is injective. Let $f'(x) = f'(y)$, then there are unique $x_1, y_1 \in A$ such that $\alpha(x_1) = x$ and $\alpha(y_1) = y$, by commutativity of the left square $\beta(f(x_1)) = f'(\alpha(x_1)) = f'(\alpha(y_1)) = \beta(f(y_1))$ so $f(x_1) = f(y_1)$ and since f is injective $x_1 = y_1$. Choose any $y \in C'$, then we find unique $y_1 \in C$ such that $\gamma(y_1) = y$, since g is surjective there is some $y_2 \in B$ such that $g(y_2) = y_1$. By the commutativity of the right square we have $g'(\beta(y_2)) = \gamma(g(y_2)) = y$, therefore g' is surjective.

It remains to prove exactness at B', let $(x,y) \in \ker(g')$ so $g'(x) = g'(y)$. If $x = y$ we are done, assume then that $x \neq y$. We find unique $x_1, y_1 \in B$ such that $\beta(x_1) = x$ and $\beta(y_1) = y$. By commutativity of the right square we get $\gamma(g(x_1)) = \gamma(g(y_1)) \implies g(x_1) = g(y_1)$. By exactness at B it follows that $(x_1,y_1) \in \mathrm{im}(f)$, since $x \neq y$ there are some $x_2, y_2 \in A$ such that $f(x_2) = x_1$ and $f(y_2) = y_1$, thus by commutativity of the left square $f'(\alpha(x_2)) = \beta(f(x_2)) = x$ and $f'(\alpha(y_2)) = \beta(f(y_2)) = y$ or $(x,y) \in \mathrm{im}(f')$.

Choose any $(x,y) \in \mathrm{im}(f')$, if $x = y$ we are done, so assume $x \neq y$, then there are some $x_1, y_1 \in A'$ such that $f'(x_1) = x$ and $f'(y_1) = y$. Since α is an iso there are unique $x_2, y_2 \in A$ so that $\alpha(x_2) = x_1$

and $\alpha(y_2) = y_1$, then by commutativity of the left square $x = \beta(f(x_2))$ and $y = \beta(f(y_2))$, by exactness at B $g(f(x_2)) = g(f(y_2))$, since $(f(x_2), f(y_2)) \in \mathrm{im}(f)$ by definition. Then we have $\gamma(g(f(x_2))) = \gamma(g(f(y_2)))$ and therefore by commutativity of the outer rectangle $g'(x) = g'(f'(\alpha(x_2))) = \gamma(g(f(x_2))) = \gamma(g(f(y_2))) = g'(f'(\alpha(y_2))) = g'(y)$ which means $(x, y) \in \ker(g')$. It follows that $\ker(g') = \mathrm{im}(f')$ as required.

Assume now that the second row is Rees exact. We can consider the following commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & A' \arrow[r, "f'"] \arrow[d, "\alpha^{-1}"] & B' \arrow[r, "g'"] \arrow[d, "\beta^{-1}"] & C' \arrow[r] \arrow[d, "\gamma^{-1}"] & 0 \\
0 \arrow[r] & A \arrow[r, "f"] & B \arrow[r, "g"] & C \arrow[r] & 0
\end{tikzcd}$$

and notice that the result follows by symmetry from the first part of the proof. □

In contrast with the category of modules, the internal hom functor isn't left exact. Indeed, consider the following counterexample.

Counterexample 0.1. Recall that $\mathbb{Z}_2$-acts are just pointed sets. Consider a short exact sequence of pointed sets

$$0 \longrightarrow \mathbb{Z}_2 \xhookrightarrow{\iota} \mathbb{Z}_3 \xrightarrow{\pi} \mathbb{Z}_3/\mathbb{Z}_2 \longrightarrow 0$$

where ι is the inclusion map and π is the canonical projection. We will show that the following sequence isn't exact at $\mathrm{Hom}(\mathbb{Z}_3, \mathbb{Z}_3)$

$$0 \longrightarrow \mathrm{Hom}(\mathbb{Z}_3, \mathbb{Z}_2) \xrightarrow{\iota_*} \mathrm{Hom}(\mathbb{Z}_3, \mathbb{Z}_3) \xrightarrow{\pi_*} \mathrm{Hom}(\mathbb{Z}_3, \mathbb{Z}_3/\mathbb{Z}_2)$$

the elements of acts in the sequence are as below

$\mathrm{Hom}(\mathbb{Z}_3, \mathbb{Z}_3)$								
0	f_1	f_2	f_3	f_4	f_5	f_6	f_7	f_8
$0 \mapsto 0$	$0 \mapsto 0$	$0 \mapsto 0$	$0 \mapsto 0$	$0 \mapsto 0$	$0 \mapsto 0$	$0 \mapsto 0$	$0 \mapsto 0$	$0 \mapsto 0$
$1 \mapsto 0$	$1 \mapsto 0$	$1 \mapsto 0$	$1 \mapsto 1$	$1 \mapsto 1$	$1 \mapsto 1$	$1 \mapsto 2$	$1 \mapsto 2$	$1 \mapsto 2$
$2 \mapsto 0$	$2 \mapsto 1$	$2 \mapsto 2$	$2 \mapsto 0$	$2 \mapsto 1$	$2 \mapsto 2$	$2 \mapsto 0$	$2 \mapsto 1$	$2 \mapsto 2$

$\mathrm{Hom}(\mathbb{Z}_3, \mathbb{Z}_2)$			
0	f_1	f_2	f_3
$0 \mapsto 0$	$0 \mapsto 0$	$0 \mapsto 0$	$0 \mapsto 0$
$1 \mapsto 0$	$1 \mapsto 0$	$1 \mapsto 1$	$1 \mapsto 1$
$2 \mapsto 0$	$2 \mapsto 1$	$2 \mapsto 0$	$2 \mapsto 1$

$\mathrm{Hom}(\mathbb{Z}_3, \mathbb{Z}_3/\mathbb{Z}_2)$			
0	f_1	f_2	f_3
$0 \mapsto [0]$	$0 \mapsto [0]$	$0 \mapsto [0]$	$0 \mapsto [0]$
$1 \mapsto [0]$	$1 \mapsto [0]$	$1 \mapsto [2]$	$1 \mapsto [2]$
$2 \mapsto [0]$	$2 \mapsto [2]$	$2 \mapsto [0]$	$2 \mapsto [2]$

we now see that $\iota \circ f_1 = f_1$, $\iota \circ f_2 = f_3$, $\iota \circ f_3 = f_4$, therefore we have

$$\mathrm{im}(\iota_*) = \{(f_1, f_3), (f_3, f_1), (f_1, f_4), (f_4, f_1), (f_3, f_4), (f_4, f_3)\} \cup \{(0, 0), \cdots, (f_8, f_8)\}$$

but $\pi \circ f_2 = f_1 = \pi \circ f_5$ hence $(f_2, f_5) \in \ker(\pi_*)$, however $(f_2, f_5) \notin \mathrm{im}(\iota_*)$.

Weaker result holds, the covariant internal hom functor does preserve chain complexes.

Proposition 0.2. *Let $(X_n, d_n)_{n\in\mathbb{Z}}$ be a chain complex and D an act, then $(Hom(D, X_n), d_{n*})$ is a chain complex.*

Proof. Choose $n \in \mathbb{Z}$ and $(f_1, f_2) \in \mathrm{im}(d_{n+1*})$, if $f_1 = f_2$ we are done, otherwise we find $h_1, h_2 \in \mathrm{Hom}(D, X_{n+1})$ such that $d_{n+1} \circ h_1 = f_1$ and $d_n \circ h_2 = f_2$, then for all $x \in D$ we have $(f_1(x), f_2(x)) \in \mathrm{im}(d_{n+1}) \subseteq \ker(d_n)$, which implies $d_n \circ f_1(x) = d_n \circ f_2(x)$ so $(f_1, f_2) \in \ker(f_{n*})$. □

Lemma 0.2 (Four lemma)**.** *Consider the following commutative diagram, whose rows are exact*

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[d, "r"] & B \arrow[r, "g"] \arrow[d, "s"] & C \arrow[r, "h"] \arrow[d, "t"] & D \arrow[d, "u"] \\
A' \arrow[r, "f'"] & B' \arrow[r, "g'"] & C' \arrow[r, "h'"] & D'
\end{tikzcd}$$

then

1. *if s, u are monomorphisms and r is an epimorphism, then t is a monomorphism.*

2. *if r, t are epimorphisms and u is a monomorphism, then s is an epimorphsm.*

Proof. To prove the first part choose $(x, y) \in \ker(t)$ and assume for a contradiction that $x \neq y$. By commutativity of the right square we have $u(h(x)) = u(h(y)) \implies h(x) = h(y)$ since u is a mono by assumption. By exactness at C and the fact that $x \neq y$ there are $x_1, y_1 \in B$ such that $g(x_1) = x$ and $g(y_1) = y$ and $x_1 \neq y_1$, then $s(x_1) \neq s(y_1)$, since s is injective by assumption, moreover we have

$$t(g(x_1)) = t(x) = t(y) = t(g(y_1))$$

hence $g'(s(x_1)) = g'(s(y_1))$, which implies $(s(x_1)), s(y_1)) \in \mathrm{im} f'$ by exactness at B', since $s(x_1) \neq s(y_1)$ there are $x_2, y_2 \in A'$ such that $f'(x_2) = s(x_1)$ and $f'(y_2) = s(y_1)$. By surjecitvity of r we find $x_3, y_3 \in A$ such that $r(x_3) = x_2$ and $r(y_3) = y_2$, then

$$s(f(x_3)) = f'(r(x_3)) = f'(x_2) = s(x_1)$$

$$s(f(y_3)) = f'(r(y_3)) = f'(y_2) = s(y_1)$$

by injectivity of s we have $f(x_3) = x_1$ and $f(y_3) = y_1$, therefore by exactness at B

$$x = g(x_1) = g(y_1) = y$$

we have reached a contradiction.

To prove the second part choose $y \in C'$, if $y = 0$ we are done, since $s(0) = 0$, assume that $y \neq 0$, then since t is surjective we find $y_1 \in C$ such that $t(y_1) = g'(y)$, then by exactness at C' and commutativity of the right square we have $u(0) = 0 = h'(g'(y)) = h'(t(y_1)) = u(h(y_1))$, hence $h(0) = 0 = h(y_1)$, since u is injective. We now have that $(0, y_1) \in \mathrm{im}(g)$ by exactness at C.

If $y_1 \neq 0$ we find $y_2 \in B$ such that $g(y_2) = y_1$, then

$$g'(s(y_2)) = t(g(y_2)) = t(y_1) = g'(y)$$

so by exactness at B' we have $(s(y_2), y) \in \mathrm{im}(f')$, if $s(y_2) = y$ we are done otherwise we find $z \in A'$ such that $f'(z) = y$ and $z_1 \in A$ such that $r(z_1) = z$, then $y = f'(r(z_1)) = s(f(z_1))$ by the commutativity of the left square.

If $y_1 = 0$, then $g'(y) = t(y_1) = 0 = g'(0)$, therefore $(y, 0) \in \mathrm{im}(f')$ by exactness at B', by assumption $y \neq 0$ so there exists $z \in A'$ such that $f'(z) = y$ and $z_1 \in A$ satisfying $r(z_1) = z$ and again we have by commutativity of the left square

$$y = f'(r(z_1)) = s(f(z_1))$$

which concludes the proof. □

Corollary 0.1 (Five lemma)**.** *Consider the following commutative diagram with exact rows*

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[d, "r"] & B \arrow[r, "g"] \arrow[d, "s"] & C \arrow[r, "h"] \arrow[d, "t"] & D \arrow[r, "k"] \arrow[d, "u"] & E \arrow[d, "v"] \\
A' \arrow[r, "f'"] & B' \arrow[r, "g'"] & C' \arrow[r, "h'"] & D' \arrow[r, "k'"] & E'
\end{tikzcd}$$

if s, u are isomorphisms, r is an epimorphism and v is a monomorphism, then t is an isomorphism.

Proof. By applying the first and second part of the lemma to our diagram we get that t is a monomorphism and an epimorphism, therefore an isomorphism, since the category of acts is balanced. □

Definition 0.5. A sequence of chain complexes

$$0 \longrightarrow (A, a) \xrightarrow{f} (B, b) \xrightarrow{g} (C, c) \longrightarrow 0$$

(where by 0 we mean $\cdots \to 0 \to 0 \to 0 \to \cdots$) is exact if for each $n \in \mathbb{Z}$ we have that

$$0 \longrightarrow A_n \xrightarrow{f_n} B_n \xrightarrow{g_n} C_n \longrightarrow 0$$

is a Rees short exact sequence.

Proposition 0.3. *If*

$$0 \longrightarrow (A,a) \xrightarrow{f} (B,b) \xrightarrow{g} (C,c) \longrightarrow 0$$

is a short exact sequence of chain complexes, then the sequence

$$h_n(A) \xrightarrow{h_n(f)} h_n(B) \xrightarrow{h_n(g)} h_n(C)$$

is Rees exact.

Proof. Choose $[x,y] \in h_n(A)$, then we have

$$(h_n(g) \circ h_n(f))([x,y]) = [g_n(f_n(x)), g_n(f_n(y))] = [0,0] \implies \operatorname{im}(h_n(f)) \subseteq \ker(h_n(g))$$

□

Proposition 0.4. *Consider a short exact sequence of chain complexes*

$$0 \longrightarrow (A,c) \xrightarrow{f} (B,d) \xrightarrow{g} (C,e) \longrightarrow 0$$

if any pair of chain complexes A,B and C is exact, so is the third.

Proof. Choose $n \in \mathbb{Z}$ and assume B and C are exact. We have the following commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & A_{n+1} \arrow[r, hook, "f_{n+1}"] \arrow[d, "a_{n+1}"] & B_{n+1} \arrow[r, two heads, "g_{n+1}"] \arrow[d, "b_{n+1}"] & C_{n+1} \arrow[r] \arrow[d, "c_{n+1}"] & 0 \\
0 \arrow[r] & A_n \arrow[r, hook, "f_n"] \arrow[d, "a_n"] & B_n \arrow[r, two heads, "g_n"] \arrow[d, "b_n"] & C_n \arrow[r] \arrow[d, "c_n"] & 0 \\
0 \arrow[r] & A_{n-1} \arrow[r, hook, "f_{n-1}"] \arrow[d, "a_{n-1}"] & B_{n-1} \arrow[r, two heads, "g_{n-1}"] \arrow[d, "b_{n-1}"] & C_{n-1} \arrow[r] \arrow[d, "c_{n-1}"] & 0 \\
0 \arrow[r] & A_{n-2} \arrow[r, hook, "f_{n-2}"] & B_{n-2} \arrow[r, two heads, "g_{n-2}"] & C_{n-2} \arrow[r] & 0
\end{tikzcd}$$

since we are dealing with chain complexes, it suffices to show $\ker(a_n) \subseteq \operatorname{im}(a_{n+1})$. Choose $x, y \in A_n$ such that $a_n(x) = a_n(y)$ and assume $x \neq y$ (otherwise we are done), then we have $b_n(f_n(x)) = f_{n-1}(a_n(x)) = f_{n-1}(a_n(y)) = b_n(f_n(y))$ which implies $(f_n(x), f_n(y)) \in \operatorname{im}(b_{n+1})$, since $f_n(x) \neq f_n(y)$ there are $x_1, y_1 \in B_{n+1}$ such that $b_{n+1}(x_1) = f_n(x)$ and $b_{n+1}(y_1) = f_n(y)$. Then we have

$$c_{n+1}(g_{n+1}(x_1)) = g_n(b_{n+1}(x_1)) = g_n(f_n(x)) = 0 = g_n(f_n(y)) = g_n(b_{n+1}(y_1)) = c_{n+1}(g_{n+1}(y_1))$$

therefore $(g_{n+1}(x_1), g_{n+1}(y_1)) \in \operatorname{im}(c_{n+2})$ by exactness.

If $g_{n+1}(x_1) = g_{n+1}(y_1)$, then $(x_1, y_1) \in \operatorname{im}(f_{n+1})$, we know $x_1 \neq y_1$, else $f_n(x) = f_n(y) \implies x = y$, a contradiction. So there are $x_2, y_2 \in A_{n+1}$ such that $f_{n+1}(x_2) = x_1$ and $f_{n+1}(y_2) = y_1$, but then

$$f_n(a_{n+1}(x_2)) = b_{n+1}(f_{n+1}(x_2)) = b_{n+1}(x_1) = f_n(x) \implies x = a_{n+1}(x_2)$$

$$f_n(a_{n+1}(y_2)) = b_{n+1}(f_{n+1}(y_2)) = b_{n+1}(y_1) = f_n(y) \implies y = a_{n+1}(y_2)$$

therefore $(x, y) \in \operatorname{im}(a_{n+1})$ as required.

If $g_{n+1}(x_1) \neq g_{n+1}(y_1)$, then since $(g_{n+1}(x_1), g_{n+1}(y_1)) \in \operatorname{im}(c_{n+2})$ we find $z_1, w_1 \in C_{n+2}$ such that $c_{n+2}(z_1) = g_{n+1}(x_1)$ and $c_{n+2}(w_1) = g_{n+1}(y_1)$. Since g_{n+2} is an epimorphism we find $z_2, w_2 \in B_{n+2}$ such that $g_{n+2}(z_2) = z_1$ and $g_{n+2}(w_2) = w_1$. Now we have

$$g_{n+1}(b_{n+2}(z_2)) = c_{n+2}(g_{n+2}(z_2)) = c_{n+2}(z_1) = g_{n+1}(x_1)$$

$$g_{n+1}(b_{n+2}(w_2)) = c_{n+2}(g_{n+2}(w_2)) = c_{n+2}(w_1) = g_{n+1}(y_1)$$

which implies that $(b_{n+2}(z_2), x_1) \in \operatorname{im}(f_{n+1})$ and $(b_{n+2}(w_2), y_1) \in \operatorname{im}(f_{n+1})$ by exactness. Now assume that $x_1 = b_{n+2}(z_2)$ then

$$f_n(x) = b_{n+1}(x_1) = b_{n+1}(b_{n+2}(z_2)) = 0 = f_n(0) \implies x = 0 = a_{n+1}(0)$$

by symmetry the same holds for y, but both cannot hold at the same time since we would have $x = y$ which would be a contradiction. Without loss of generality we will only consider the case when $x_1 \neq b_{n+2}(z_2)$ and $y = 0 = a_{n+1}(0)$, then since $(x_1, b_{n+2}(z_2)) \in \mathrm{im}(f_{n+1})$ we find $x_2 \in A_{n+1}$ such that $f_{n+1}(x_2) = x_1$ then

$$f_n(a_{n+1}(x_2)) = b_{n+1}(f_{n+1}(x_2)) = b_{n+1}(x_1) = f_n(x) \implies a_{n+1}(z_1) = x$$

Now assume that A and B are exact. Choose $x, y \in \ker(c_n)$, $x \neq y$, find $x_1, y_1 \in B_n$ such that $g_n(x_1) = x$,$g_n(y_1) = y$, then $g_{n-1}(b_n(x_1)) = c_n(x) = c_n(y) = g_{n-1}(b_n(y_1))$ which implies $(b_n(x_1), b_n(y_1))\mathrm{im} f_{n-1}$.

If $b_n(x_1) = b_n(y_1)$, then $(x_1, y_1) \in \mathrm{im}(b_{n+1})$ (note that we have $x_1 \neq y_1$, because $x \neq y$) so we find $b, c \in B_{n+1}$ such that $b_{n+1}(b) = x_1$ and $b_{n+1}(c) = y_1$, then

$$x = g_n(x_1) = g_n(b_{n+1}(b)) = c_{n+1}(g_{n+1}(b))$$

$$y = g_n(y_1) = g_n(b_{n+1}(c)) = c_{n+1}(g_{n+1}(c))$$

therefore $(x, y) \in \mathrm{im}(c_{n+1})$.

If $b_n(x_1) \neq b_n(y_1)$, then since $(b_n(x_1), b_n(y_1)) \in \mathrm{im}(f_{n-1})$ we find $x_2, y_2 \in A_{n-1}$ such that $f_{n-1}(x_2) = b_n(x_1)$ and $f_{n-1}(y_2) = b_n(y_1)$. Now we compute

$$f_{n-2}(a_{n-1}(x_2)) = b_{n-1}(b_n(x_1)) = 0 = b_{n-1}(b_n(y_1)) = f_{n-2}(a_{n-1}(y_2))$$

hence $a_{n-1}(x_2) = a_{n-1}(y_2)$ and by exactness $(x_2, y_2) \in \mathrm{im}(a_n)$. If $x_2 = y_2$ we would have $b_n(x_1) = b_n(y_1)$ which contradicts our assumption so we find $x_3, y_3 \in A_n$ such that $a_n(x_3) = x_2$ and $a_n(y_3) = y_2$ then $b_n(f_n(x_3)) = f_{n-1}(a_n(x_3)) = b_n(x_1)$ and similarly $b_n(f_n(y_3)) = b_n(y_1)$. So by exactness $(f_n(x_3), x_1) \in \mathrm{im}(b_{n+1})$ and $(f_n(y_3), y_1) \in \mathrm{im}(b_{n+1})$. From now on we will only consider $(x_1, f_n(x_3))$ because the same reasoning applies to the other pair. If $f_n(x_3) = x_1$, then $x = g_n(x_1) = g_n(f_n(x_3)) = 0 = c_{n+1}(0)$. If $f_n(x_3) \neq x_1$, then we find $z_1 \in B_{n+1}$ such that $x_1 = b_{n+1}(z_1)$, then

$$x = g_n(x_1) = g_n(b_{n+1}(z_1)) = c_{n+1}(g_{n+1}(z_1))$$

in either case x in in the image of c_{n+1} as required.

Finally assume that A and C are exact. Choose $(x, y) \in \ker(b_n)$,$x \neq y$, then

$$c_n(g_n(x)) = g_{n-1}(b_n(x)) = g_{n-1}(b_n(y)) = c_n(g_n(y)) \implies (g_n(x), g_n(y)) \in \mathrm{im}(c_{n+1})$$

If $g_n(x) = g_n(y)$, then $(x, y) \in \mathrm{im}(f_n)$, we find $x_1, y_1 \in A_n$ such that $f_n(x_1) = x$ and $f_n(y_1) = y$, then $x_1 \neq y_1$ since $x \neq y$ and we have

$$f_{n-1}(a_n(x_1)) = b_n(f_n(x_1)) = b_n(x) = b_n(y) = f_{n-1}(a_n(y_1)) \implies a_n(x_1) = a_n(y_1)$$

hence by exactness we find $x_2, y_2 \in A_{n+1}$ such that $a_{n+1}(x_2) = x_1$ and $A_{n+1}(y_2) = y_1$, then $x = f_n(a_{n+1}(x_2)) = b_{n+1}(f_{n+1}(x_2))$ and similarly $y = b_{n+1}(f_{n+1}(y_2))$, thus $(x, y) \in \mathrm{im}(b_{n+1})$.

If $g_n(x) \neq g_n(y)$, then we find $z_1, w_1 \in C_{n+1}$ such that $c_{n+1}(z_1) = g_n(x)$, $c_{n+1}(w_1) = g_n(y)$. Next we find $z_2, w_2 \in B_{n+1}$ such that $g_{n+1}(z_2) = z_1$, $g_{n+1}(w_2) = w_1$, then $g_n(b_{n+1}(z_2)) = c_{n+1}(g_{n+1}(z_2)) = g_n(x)$ and $g_n(b_{n+1}(w_2)) = c_{n+1}(g_{n+1}(w_2)) = g_n(y)$, thus $(b_{n+1}(z_2), x) \in \mathrm{im}(f_n)$,$(b_{n+1}(w_2), y) \in \mathrm{im}(f_n)$.

If $x = b_{n+1}(z_2)$ and $y = b_{n+1}(w_2)$ we are done. Assume first that $x \neq b_{n+1}(z_2)$ and $y = b_{n+1}(w_2)$. We find $r_1 \in A_n$ such that $f_n(r_1) = x$, then

$$f_{n-1}(a_n(r_1)) = b_n(f_n(r_1)) = b_n(x) = b_n(y) = b_n(b_{n+1}(w_2)) = 0 = f_{n-1}(0)$$

The penultimate equality follows from the fact that (B, b) is a chain complex. Since f_{n-1} is a monomorphism we have $a_n(r_1) = 0 = a_n(0)$, so $(r_1, 0) \in \mathrm{im}(a_{n+1})$. If $r_1 = 0$, then $x = 0 = b_{n+1}(0)$. Assume $r_1 \neq 0$, then we find $r_2 \in A_{n+1}$ such that $a_{n+1}(r_2) = r_1$, then $x = f_n(a_{n+1}(r_2)) = b_{n+1}(f_{n+1}(r_2))$ and we are done.

The case $x = b_{n+1}(z_2)$ and $y \neq b_{n+1}(w_2)$ resolves in the same way as the last one, by symmetry.

Assume $x \neq b_{n+1}(z_2)$ and $y \neq b_{n+1}(w_2)$, then we find $r_1, s_1 \in A_n$ such that $f_n(r_1) = x$,$f_n(s_1) = y$. We have

$$f_{n-1}(a_n(r_1)) = b_n(f_n(r_1)) = b_n(x) = b_n(y) = b_n(f_n(s_1)) = f_{n-1}(a_n(s_1)) \implies a_n(r_1) = a_n(s_1)$$

We cannot have $r_1 = s_1$, since it would contradict our assumption $x \neq y$. So we find $r_2, s_2 \in A_{n+1}$ such that $a_{n+1}(r_2) = r_1$,$a_{n+1}(s_2) = s_1$, then

$$x = f_n(r_1) = f_n(a_{n+1}(r_2)) = b_{n+1}(f_{n+1}(r_2))$$

$$y = f_n(s_1) = f_n(a_{n+1}(s_2)) = b_{n+1}(f_{n+1}(s_2))$$

which concludes the proof. $\square$

Corollary 0.2 (3x3 lemma)**.** *Consider the following commutative diagram, whose rows are Rees exact*

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & A_3 \arrow[r, "f_3"] \arrow[d, "r_3"] & B_3 \arrow[r, "g_3"] \arrow[d, "r_2"] & C_3 \arrow[r] \arrow[d, "r_1"] & 0 \\
0 \arrow[r] & A_2 \arrow[r, "f_2"] \arrow[d, "s_3"] & B_2 \arrow[r, "g_2"] \arrow[d, "s_2"] & C_2 \arrow[r] \arrow[d, "s_1"] & 0 \\
0 \arrow[r] & A_1 \arrow[r, "f_1"] \arrow[d] & B_1 \arrow[r, "g_1"] \arrow[d] & C_1 \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

1. *If the first and the second columns are exact, then so is the third.*
2. *If the second and the third columns are exact, then so is the first.*
3. *If the first and the third columns are exact and additionaly $s_2 \circ r_2 = 0$, then so is the second.*

Proof. For the first part we will have to show, that the sequence

$$\cdots \longrightarrow 0 \longrightarrow C_3 \xrightarrow{r_1} C_2 \xrightarrow{s_1} C_1 \longrightarrow 0 \longrightarrow \cdots$$

is a chain complex. It suffices to prove that $\mathrm{im}(r_1) \subseteq \ker(s_1)$. We have to show $s_1 \circ r_1 = 0$, since g_3 is epic we have

$$s_1 \circ r_1 \circ g_3 = s_1 \circ g_2 \circ r_2 = g_1 \circ s_2 \circ s_1 = g_1 \circ 0 = 0 = 0 \circ g_3 \implies s_1 \circ r_1 = 0$$

Now applying the previous proposition to the evident chain complex yields the statement.

Assume now that the second and the third columns are exact. We again only have to show that $s_3 \circ r_3 = 0$. We have

$$f_1 \circ s_3 \circ r_3 = s_2 \circ f_2 \circ r_3 = s_2 \circ r_2 \circ f_3 = 0 \circ f_3 = 0 = f_1 \circ 0 \implies s_3 \circ r_3 = 0$$

For the third part we can just directly apply the previous proposition. This completes the proof. $\square$